Journal of Systems Architecture 57 (2011) 869–885

Contents lists available at ScienceDirect

Journal of Systems Architecture

journal homepage: www.elsevier.com/locate/sysarc

Weapon classification and shooter localization using distributed multichannel acoustic sensors

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ARTICLE INFO

Article history:
Available online 19 May 2011

Keywords:
Sensor networks
Data fusion
Acoustic source localization
Weapon classification
Caliber estimation

ABSTRACT

A wireless sensor network-based wearable countersniper system prototype is presented. The sensor board is connected to a small helmet-mounted microphone array that uses time of arrival (ToA) estimates of the ballistic shockwave and the muzzle blast to compute the angle of arrival (AoA) of both acoustic events. A low-power radio is used to form an ad-hoc multihop network that shares the detections among the nodes. Utilizing all available ToA and AoA data, a novel sensor fusion algorithm then estimates the shooter position, bullet trajectory, miss distance, caliber, and weapon type. A single sensor relying only on its own detections is able determine the shooter position when both the shockwave and the muzzle blast are detected by at least three microphones each. Even with just one shockwave and one muzzle blast detection, the miss distance and range can be accurately estimated by a single sensor. The system has been tested multiple times at the US Army Aberdeen Test Center and the Nashville Police Academy. The demonstrated performance is 1-degree trajectory precision, over 95% caliber estimation accuracy, and close to 100% weapon estimation accuracy for 4 out of the 6 guns tested.

1. Introduction

Military applications have been some of the main driving forces behind embedded systems research. As a prime example, countersniper systems have experienced rapid innovation in the last decade. They have progressed from the bulky and expensive systems of yesterday to the personal, wearable systems that have started to appear recently. However, these systems are still standalone; we have yet to see a networked countersniper system deployed. The research results described in this paper detail the many advantages that widely distributed sensing brings to acoustic shooter localization.

So far, the only networked system in the literature is PinPtr, a wireless sensor network (WSN) based prototype we created in 2003 [18]. In contrast to the traditional approach of using one or a couple of expensive microphone arrays, PinPtr is based on a large number of inexpensive single-channel sensors that communicate with each other via a low-power radio. The sensor nodes detect the muzzle blast of the shot and the ballistic shockwave generated by the supersonic projectile and send their times of arrival (TOA) to a laptop computer that fuses the observations and displays the estimated shooter location. The widely distributed sensing and unique sensor fusion algorithm [12] enable the system to eliminate erroneous observations due to non line of sight (LOS) conditions (i.e. echoes) and to resolve multiple simultaneous shots. The demonstrated accuracy of the system is about 1 m on average in 3D for shots originating within or near the sensor network. For longer range shots, 1 degree bearing precision for both azimuth and elevation and 10% accuracy in range estimation have been achieved in various US Army Military Operations in Urban Terrain (MOUT) facilities.

While the performance of PinPtr is very good, it has a number of disadvantages. First, the number of sensors required for accurate localization and wide coverage is relatively high, especially in urban environments. Second, the reusability of the sensors is limited because it is not reasonable to expect the soldiers to gather the sensor nodes once they are not needed in a given area. These observations led to the development of a soldier-wearable version of PinPtr. The obvious advantage is that the system goes wherever the soldiers do, so there is constant protection provided. The main difference between our prototype and commercially available systems [2,8] is that those systems comprise standalone devices, while our approach networks the individual sensor nodes together to form a widely distributed sensor field. The achievable precision is consequently much higher, and the approach provides new capabilities, such as caliber estimation and weapon classification.

A wearable countersniper system presents significant challenges. The number of sensors in a given area is significantly

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doi:10.1016/j.sysarc.2011.04.003

smaller than with PinPtr. Also, the system needs to work for a single isolated soldier as well. Hence, each sensor node needs to be equipped with multiple microphones, forming a small array. This enables angle of arrival (AoA) detection, but as a consequence, not only the position, but also the orientation of the sensor node needs to be tracked.

The rest of the paper is organized as follows. We start by reviewing the relevant literature. Sections 3–6 describe the hardware and software architecture, including the custom sensor board equipped with a small microphone array and connected to a low power radio node [3]. The sensor fusion technique that estimates the shooter position, projectile trajectory, miss distance, caliber, and weapon type is described in Section 7. Section 8 explains how a single device operates in standalone mode when there is no network connectivity. Finally, in Section 9, we conclude by presenting sensor fusion results using prerecorded shockwave and muzzle blast arrival times of live fire shots, significantly improving the localization rate and weapon classification accuracy of the initial system evaluated at the US Army Aberdeen Test Center in 2006 as reported in [20].

2. Related work

Research on acoustic gunshot detection has a long history. Fansler studied the ideal muzzle blast pressure wave in the near field without contamination from echoes or propagation effects [9]. For longer ranges, Stoughton [19] took measurements of ballistic shockwaves using calibrated pressure transducers at known locations, measured bullet speeds, and miss distances of 3–55 m for 5.56 mm and 7.62 mm projectiles.

A related area of research is the signal processing of gunfire acoustics for the robust detection of small caliber acoustic shockwaves and muzzle blasts. Chacon-Rodriguez presents an evaluation of a number of proposed algorithms [4]. In a WSN-based solution, such as the one presented in this work, the severe resource constraints on the battery-operated sensors nodes, the limited communication bandwidth, and the real-time requirements render these approaches infeasible. Our simple, time domain-based technique fits well within these constraints and performs satisfactorily.

As shockwave length is directly related to bullet characteristics and the edges of a shockwave are typically well defined, Sadler [16] demonstrates how the length of a shockwave can be used along with Whitham's equations [22] to estimate the caliber of a projectile.

There are many systems in the literature, some even available commercially, that localize the origin of a shot based on detected acoustic signals. This localization is based on the ToA of the shockwave and/or the muzzle blast at multiple microphones. Most systems are centralized consisting of one or a few microphone arrays with wired connections to a central processing unit. The microphones within the array have precisely-known separations, enabling accurate AoA detection. An early such system was Bullet Ears [7]. A few years ago, BBN introduced the vehicle-mounted BOOMERANG that is being used in Iraq and Afghanistan [1]. The most recent system that the US military deployed is made by QinetiQ [8]. It is a wearable device consisting of a pod with four shoulder-worn microphones and a second hand-held component that displays the location of the shots. The typical bearing precisions provided by these systems are 10 or more degrees. Also, the problem with these and similar centralized systems is the need for LOS. A WSN-based solution has the advantage of widely-distributed sensing for better coverage and accuracy, multipath elimination, and multiple simultaneous shot resolution [12]. This is especially important for operation in acoustically reverberant urban areas.

Recently, other researchers have started to explore the domain of networked sensing for shooter localization. Damarla et al. presents an approach that does not require the distributed sensor to be time synchronized [5]. The technique relies only on the time difference of arrival (TDoA) of the shockwave and muzzle blast on the same nodes, but assumes a known bullet speed. Early results are based on data collected on high-quality instrumentation microphones and offline processing. Lindgren et al. presents a similar approach [13]. Considering that time synchronization is available on most WSN platforms with little overhead, it is not clear that the tradeoffs these techniques provide are relevant or necessary.

3. Approach

3.1. The physical phenomenon

When a typical rifle is fired, two distinct acoustic phenomena are observable: a ballistic shockwave and a muzzle blast. In many cases, they are both observable by the human ear, resulting in two distinct high-energy sound pulses.

3.1.1. Acoustic shockwave

A shockwave is generated by the supersonic projectile; the bullet effectively compresses the air in front of it, creating a sonic boom. The shape of the wavefront is conical, with the axis of the cone being the trajectory of the bullet. The aperture of the cone depends on the Mach number, i.e. the ratio of the bullet speed to the speed of sound. No shockwave is observable for subsonic projectiles (many handguns and shotguns), when this ratio is less than 1. For supersonic bullets, the aperture of the cone decreases with increasing bullet speed.

The wavefront of the acoustic shockwave is not a perfect conical surface for two reasons. First, the trajectory is not a straight line, but a parabolic curve (due to the force of gravity acting on the bullet), perturbed by other factors such as wind, changes in air pressure, etc. Second, the bullet is gradually decelerating after leaving the muzzle due to friction with the air, which results in the aperture of the cone-like surface not being constant, but rather increasing as the bullet loses speed.

When the acoustic shockwave hits a microphone, a characteristic signal shape resembling a capital letter N can be observed in the time domain. The rise time at both the start and end of the signal is very fast, under 1 μs. The length is typically a few hundred microseconds, which depends on the caliber and the miss distance – the distance between the trajectory and the microphone – and other less significant factors.

3.1.2. Muzzle blast

The second characteristic acoustic phenomenon produced by firing a weapon is the muzzle blast. It is a high-energy acoustic signal originating from the muzzle. The muzzle blast generates a spherical wavefront, propagating at the speed of sound, with the center of the sphere being the location of the muzzle of the gun.

3.2. The sensing approach

Our sensing approach is based on a four-channel acoustic sensor board, connected to an array of four microphones that are located 2–4″ from one another. The sensor board detects shockwave and muzzle blast events, measuring their time of arrival (ToA). If at least three acoustic channels detect the same event, its angle of arrival (AoA) can also be computed. Note that the AoA is ambiguous if exactly three detections are available; however, this ambiguity can often be resolved easily by excluding cases that are geometrically impossible (e.g. shot coming from underground).

If both the shockwave and muzzle blast AoA are available, a simple analytical solution gives the shooter location as shown in Section 7.

Since the sensor itself is compact in the soldier-wearable version of the platform, the aperture of the microphone array is small. This has two important consequences. First, the small microphone separation (2–4″) dictates that a high sampling rate is needed to precisely measure time difference of arrival (TDoA) values, which are necessary for AoA computation. Assuming a circular microphone arrangement with a maximum of 4″ of separation between neighbors, the largest possible TDoA of about 250 μs can occur along the diagonals. To achieve a high angular resolution, a sampling rate of 1 MSPS is reasonable. Since the shockwave length is in the same range, the relatively high sampling rate enables its precise estimation as well.

The second implication of employing small aperture acoustic arrays is that estimating the curvature of the shockwave and muzzle blast wavefronts is not possible with a single four channel sensor – not even at this high sampling rate. Because of this, a single sensor cannot estimate the projectile trajectory (although it can determine the shooter location). In fact, an infinite number of trajectory-bullet speed pairs satisfy the observations.

Relaxing the problem by making further assumptions allows us to do more with fewer than three detections. Relying on assumptions about the weapon caliber (enemy forces commonly use 7.62 mm ammunition), the length of the shockwave can be used to estimate the miss distance, i.e. the distance of the particular microphone from the trajectory. If there is at least one microphone that detected both a shockwave and a muzzle blast, it is further possible to compute the range of the shooter from the sensor.

Fusing detections from a network of spatially-distributed sensors that make up a wireless large-aperture acoustic sensor array makes it possible to improve the range and location detections of standalone sensors, as well as to compute the bullet trajectory, speed, caliber, and to infer what type of weapon was used. If the bullet passes through the sensor field, three shockwave AoAs are sufficient to compute the trajectory with an analytical formula. Once a trajectory estimate is available, we can refine the previously computed shooter location (e.g. it must be a point on the trajectory). Furthermore, the miss distances can now easily be computed (distance of the sensor from the estimated trajectory), which allows for caliber estimation since the measured shockwave length is a function of the miss distance and the caliber.

3.3. Sensor hardware

Clearly, precise position and orientation of the microphone array is essential for the sensor fusion to work. The orientation of the microphone array at the time of detection is provided by a 3-axis digital compass. Currently, the system assumes that the soldier's PDA is GPS-capable, and it does not provide self-localization services itself. However, the accuracy of GPS is only a few meters, degrading the overall accuracy of the system. Refer to Section 9 for an analysis. The latest generation sensor board features a Texas Instruments CC-1000 radio, enabling the high-precision radio interferometric self-localization approach we have developed separately [11]. However, we leave the integration of the two technologies for future work.

The onboard Bluetooth module of the sensor board enables communication with the soldier's PDA or laptop computer. A wired USB connection is also available. The sensor fusion algorithm and the user interface can access data through any of these channels. The sensor boards are also connected to COTS MICAz motes, and they share their AoA and ToA measurements, as well as their own locations and orientations, with each other using a multihop routing service [14]. The sensor fusion algorithm then estimates the trajectory, range, caliber, and weapon type based on all the measurements available.

4. Hardware

Earlier versions of our sensor network-based countersniper system were built upon the MICA product line [15] from UC Berkeley and Crossbow, Inc. While it is indeed possible to implement simple time-domain signal processing algorithms on such microcontroller-based platforms, they lack the required computational power and do not support the concurrent megahertz-scale sampling rates on multiple channels that are essential to carry out shockwave and muzzle blast AoA measurements with sufficient precision.

By using FPGAs, however, this problem becomes tractable: our 3rd generation acoustic sensor board is designed to be used with the MICA family of motes, and it is comparable in size to the mote itself (see Fig. 1a). The sensor board is built around the mid-range Xilinx XC3S1000 FPGA chip with various standard peripheral IP cores, multiple soft processor cores, and custom logic for the acoustic detectors (Fig. 1b).

4.1. Sensors

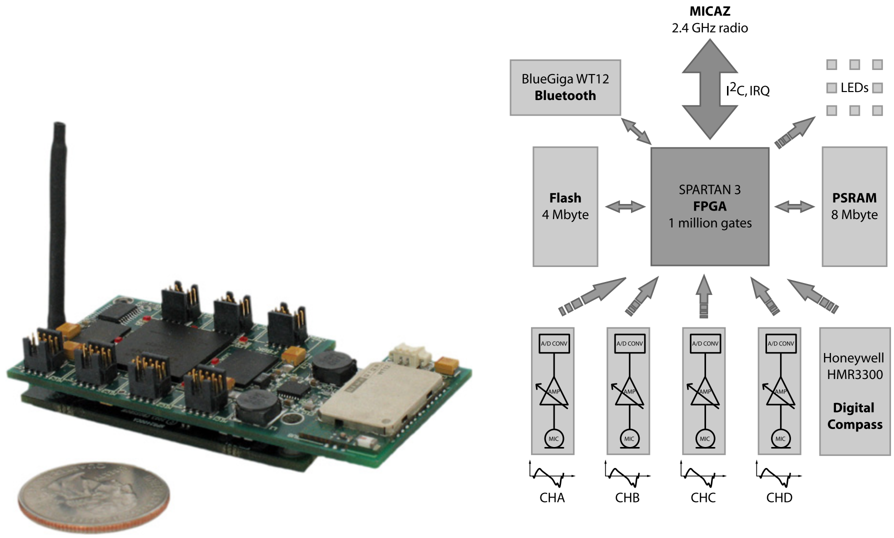
To allow for local AoA detections, the board supports four independent audio channels. For greater flexibility and also to reduce the lengths of analog signal paths, the electret microphone (Panasonic WM-64PNT), amplifiers with controllable gain (30–60 dB), and the 12-bit serial ADC (Analog Devices AD7476) reside on separate tiny boards which are connected to the main sensor board with ribbon cables carrying high-speed digital data. With such partitioning, it is also possible to use different analog front ends for each digital audio channel (to diversify gain, sampling rates, etc.); however, we opted not to do so later in this project.

For soldier-wearable applications, the position and orientation of the microphone array may change between shockwave and muzzle blast detections corresponding to the same shot. This must be detected by the board and compensated for in the sensor fusion. Therefore, the sensor board is equipped with an integrated Honeywell HMR3300 digital compass module that provides heading, pitch, and roll information with 1 degree accuracy.

4.2. Interfaces

Several communication interfaces are provided by the sensor board. To establish a data link between the sensor board and a PC or PDA, an RS232 port and a Bluetooth (BlueGiga WT12) virtual UART port are provided. Interfacing with the mote is carried out over the MICA bus: a 51-pin connector, supported by several different sensor platforms from UC Berkeley, Crossbow, Inc., ETH Zurich, etc. The sensor board uses several GPIO lines on the MICA bus (for signaling and precise time synchronization), as well as the I^2C bus for data transfer.

For in-system programming and debugging, a dedicated JTAG connector is provided. The sensor board also supports full-speed USB transfers (with custom USB dongles) for uploading recorded audio samples to the PC. The sensor board is equipped with 4 MB onboard Flash memory and 8 MB PSRAM, which allow for storing raw samples of several acoustic events. This is particularly helpful for debugging and for data acquisition, e.g. to build libraries of various acoustic signatures and to help refine the detection cores offline. These memory blocks are also used to store program images for soft-core microcontrollers synthesized in the FPGA fabric, as well as configuration data for the detection algorithm.



(a) (b)

Fig. 1. (a) Sensor hardware prototype (b) and its block diagram.

4.3. Power

Since Xilinx FPGAs have complex voltage requirements, we opted for dedicated power supply circuitry on the sensor board, which also powers the mote over the MICA bus. We used a quad pack of rechargeable AA batteries as the power source (although any other configuration is viable that meets the voltage requirements). The FPGA core (1.2 V) and I/O (3.3 V) voltages are generated by a highly efficient buck switching regulator. The FPGA configuration (2.5 V) and a separate 3.3 V power net are fed by low-current LDOs, the latter of which provides independent power to the mote and to the Bluetooth radio. To allow for power saving, the voltage regulators can be independently turned on or off from the mote or through the Bluetooth radio.

4.4. Packaging

We built several prototype sensor boards for the shooter localization system described in this paper. Some of the nodes were mounted on kevlar military helmets, with the microphone boards spaced about 6″ apart on the helmet's surface, as shown in Fig. 2a. We packaged the rest of the nodes in plastic enclosures which offered 2–4″ microphone separation. The latest solution provides more rugged and weatherproof packaging, as shown in Fig. 2b.

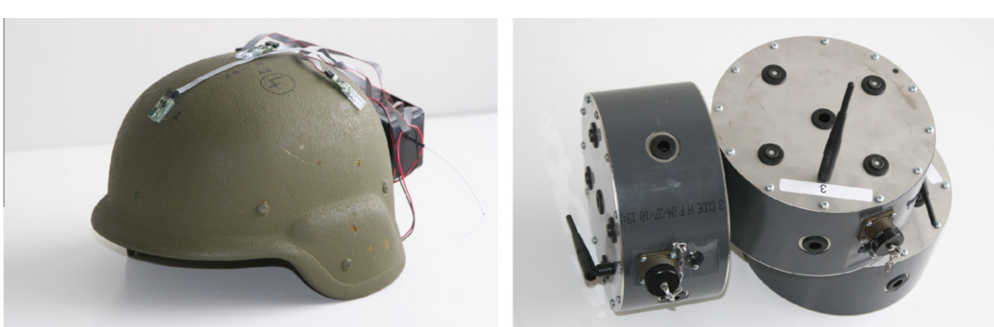
5. Software architecture

From the software perspective, our shooter localization system encompasses a broad spectrum of computing paradigms, as shown in Fig. 3. This diversity is primarily dictated by the particular hardware platforms (FPGA, low-power microcontroller, PDA, PC). While each of these hardware platforms suit their domain-specific tasks extremely well, their different models of computation, communication interfaces, and clock domains make software development and integration a great challenge.

5.1. FPGA software

The acoustic sensor board is built around the Xilinx XC3S1000 FPGA, which is a mid-range model from Xilinx's product line. The FPGA is responsible for the following tasks:

- High-speed data acquisition from the audio channels.
- Event detection (shockwave and muzzle blast) and feature extraction (e.g. shockwave length) from the sample streams.
- Time synchronization.



(a) (b)

Fig. 2. (a) Sensor prototypes mounted on a kevlar helmet (b) and in a rugged enclosure.

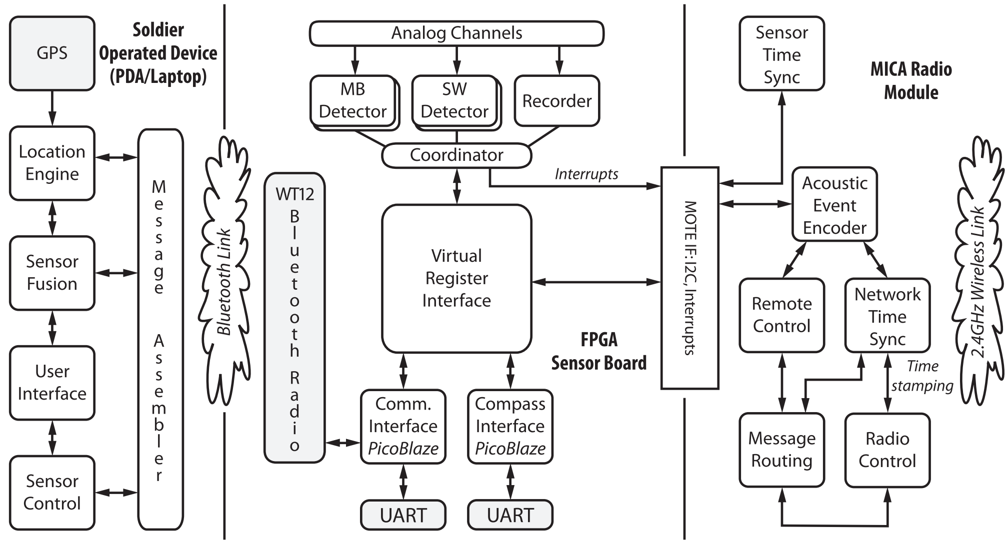


Fig. 3. Software architecture.

- Communicating detection results to subsystems (Bluetooth, mote, etc.)
- Providing an online configuration service that allows for tuning detection parameters.
- Recording sample streams and downloading them to a PC for debugging/evaluation purposes.

Some of these tasks lend themselves well to an FPGA-based implementation. For instance, high-speed data acquisition and parallel processing of the sample streams are especially well-suited for an FPGA – not even high-end DSPs are capable of carrying out such parallel tasks at a 1 MSPS sampling rate. Another advantage of FPGAs is the deterministic timing they offer. Therefore, we implemented the time-critical and/or computationally expensive tasks in VHDL as IP cores. However, implementing higher-level protocols (configuration interface, accessing the register file, UART communication with the Bluetooth module or digital compass) is typically cumbersome in a hardware description language. Since these are typically not timing-critical tasks, nor are they so computationally expensive that they warrant an efficient, HDL-based implementation, we opted to use a synthesized soft processor core in the FPGA fabric for this purpose.

5.1.1. Virtual register file

Integration of functional blocks within the FPGA is achieved by using a virtual register file, as shown in Fig. 3. The register file contains the address decoding logic, registers for storing parameter values and detection results, and point-to-point data buses to and from the peripherals. Aside from serving as a "communication hub" for IP cores, the registers are also available through the UART link and through a command line interface over the USB link and the I^2C bus. Since the various low-level interfaces are hidden behind the register file, one can easily add or modify functionality without incurring changes to any of the above interface implementations.

5.1.2. Data acquisition

Since the data acquisition and signal processing paths exhibit clear symmetry, the corresponding VHDL functional blocks are instantiated four times and execute independently in parallel. The parallel signal paths "meet" only just before the register file. All four analog channels are driven by a serial A/D core (for providing a 20 MHz serial clock and shifting in 8-bit data samples at 1 MSPS) and a digital potentiometer driver for setting the required gain. Within the FPGA fabric, each channel has a dedicated shockwave and muzzle blast detector, the details of which are described in Section 6. A central register file is used to store runtime parameters for the detectors, as well as to save detection results. The coordinator core constantly monitors the output of the detectors and generates an event on one of the general purpose IO pins of the FPGA when detection data is available for the mote to download. The I/O line is connected to an interrupt pin on the mote's microcontroller: a detection event from the FPGA manifests as an interrupt request on the mote. For efficiency reasons, the FPGA does not signal each shockwave or muzzle blast detection on different channels as single events. The state of the I/O line is changed either if all four channels detected both events, or if there is a timeout after a subset of all possible events are reported by the detectors. This way, the mote will be able to download detection results in batches, decreasing the interrupt load and protocol overhead.

While the sample-recording functionality of the sensor board is not used in the final deployment, it is essential for development purposes, such as fine tuning the detection algorithms or refining parameter values of the weapon classifier. This component receives the samples from all channels and stores them in circular buffers in the PSRAM device. The recorder component can be suspended either through the command-line configuration interface provided over the USB link, or with some delay after the signal amplitude on any of the channels exceeds a predefined threshold. The buffer length, delay, sampling rate, energy threshold, etc. are runtime configurable over the command line interface. After recording is suspended, the contents of the sample buffers are available for download over the USB link. Note that the operation of the recorder core is independent of other signal processing modules, such as the detector cores; therefore, it allows for in-place debugging of the live, fully operational system.

The FPGA cores are implemented in VHDL, while the PicoBlaze programs are written in assembly. The complete configuration occupies 40% of the resources (slices) of the FPGA, and the maximum clock speed is 30 MHz, which is comfortably higher than the actual speed at which the device is run (20 MHz).

5.1.3. Interfaces

The mote interface consists of several direct general purpose I/O lines used for signaling, and an I^2C slave IP core with a thin adaptation layer that provides a data and address bus abstraction over the I^2C protocol. The maximum bandwidth of the I^2C link is 100 Kbps.

Multiple UART links – one that connects to the onboard Bluetooth module, one for the digital compass, and one that provides a direct wired link to an onboard connector – are implemented as instantiations of HDL UART cores. The control, status, and data registers of the UART modules are available through the register file. The higher-level protocols over the UART lines are implemented in software, running on a PicoBlaze soft-core microcontroller. These include (1) parsing compass readings and storing them in the register file, and (2) implementation of an interactive command-line interface (for testing and debug purposes).

Two full-speed USB links are available on the sensor board, provided by the FT245R USB device controller. One of them is tied to the recorder block in parallel FIFO mode and is used to download recorded samples to the PC. The second USB link is configured as Serial-over-USB and provides the command-line interface over the USB connection.

The RAM interface, connected to the onboard 8 MB PSRAM, is an IP core that implements data read/write cycles with the correct timing required by the memory chip. Functionally, the IP core interfacing with the memory chip is connected directly to the recorder IP core. Together with the full-speed USB link, they provide an effective bandwidth of 1 MB/s, which is sufficient to stream recorded audio data to the PC.

5.2. Mote software

The main task of the MICAz motes is to distribute detection results in the sensor network. Since detection results consist of signal features as well as detection timestamps, the motes carry out time synchronization, as well as multihop routing.

We use the TinyOS operating system on the motes. Under TinyOS, applications are implemented in the nesC [10] programming language. The application code, integrated with the OS with its 3 kB RAM and 28 kB program space (ROM) requirement, easily fits on the MICAz motes.

5.2.1. Time synchronization with the FPGA

When the FPGA signals the availability of new detection results to the mote through a dedicated interrupt line, the mote reads the detection timestamps from the register file through the I^2C interface. The FPGA's and the mote's clock domains are not synchronized: in fact, they do not even have the same clock rates. The FPGA's clock is ticking at 1 MHz, while the mote has only a 921.6 kHz counter. Since the detection timestamps are given in the FPGA's clock domain, we need to translate them into the mote's clock domain. The mote achieves this by periodically requesting the capture of the FPGA's current timer value through a dedicated GPIO line and reading the captured value from the register file through the I^2C interface. Based on the current and previous readings and the corresponding mote's local timestamps, the mote can calculate and maintain the scaling factor and offset between the two time domains. The offset and scaling factor are then applied to the downloaded timestamps to map them to the mote's time.

5.2.2. Multihop routing

Detection messages are disseminated in the sensor network via controlled flooding. We use the directed flood-routing framework (DFRF) [14] in our application. DFRF is a policy-based routing framework, where policies define rules for accepting, discarding, storing, and resending messages. The application can change the policies at runtime, i.e. switching to a gradient converge-cast policy to collect detection data at a designated sink, if this is what the application scenario requires.

5.2.3. Time synchronization in the ad-hoc network

Detection timestamps must be converted to a common time base in order for the sensor fusion to work. Instead of maintaining a global virtual time in the network, we chose to implement post-facto time synchronization. In particular, we use the routing integrated time synchronization (RITS) [17] approach. RITS relies on very accurate MAC-layer time-stamping to embed the cumulative delay in the radio message that has been accrued enroute to the destination since the time of the event detection. Namely, each node along the routing path measures the time the message spent there, and adds this value to the number of the accumulated delay field in the payload. Every receiving node can subtract the delay from its current time to obtain the detection time in its local time reference. The per-hop accuracy of RITS on the MICAz motes is better than 10 μs, which is more than adequate for this particular system.

5.3. Sensor fusion software

The sensor fusion module, running on a PC or PDA, is linked to the sensor nodes with serial-over-Bluetooth or serial-over-USB lines. The primary task of the software, implemented in Java to allow for portability, is to compute the shooter's location, bullet trajectory, and weapon type from the available data: shockwave and muzzle blast detections, compass headings, and GPS coordinates (available on the PDA). The software has a visualizer component, which is platform specific (with different implementations for the PC and for the PDA), that displays sensor fusion results on the map.

6. Detection algorithm

The shockwave and muzzle blast detection algorithms have a number of unique requirements that distinguish them from regular audio applications. Both the shockwave and muzzle blast signals are short and transient in nature. Events may come frequently, e.g. when an automatic weapon is fired, or when the geometry is such that the shockwave and muzzle blast arrival times are less than a millisecond away. The detection algorithm must therefore run in near real-time, settling soon after the signals reach the microphone. Also, AoA computation with relatively small microphone separations mandates high sampling rates and processing multiple audio channels in parallel.

Depending on the sensor's distance from the trajectory/muzzle, both the shockwave and muzzle blast can manifest as very high-energy signals. Therefore, from the hardware point of view, low-cost electret microphones, which are designed with consumer electronics applications in mind, are not particularly suitable for this purpose. While mechanical damping does mitigate the problem of microphone saturation, this approach also has side effects, as it alters the microphone's frequency response. Hence, the detection algorithm must be robust enough to handle nonlinear distortion and saturation and recovery (transitory oscillation).

Considering the unique application requirements, as well as the constraints and capabilities of the FPGAs available on the market, we chose to implement the detection algorithms in the time domain with state machine logic. Instead of using simple energy threshold-based detectors that might not be able to differentiate between shockwave and muzzle blast events, the detectors we developed are capable of identifying the features of interest in the input signal. While in the general case, frequency-domain signal analysis may provide more robust results, there are two important reasons why we decided against using them. First, frequency-domain based calculations are computationally expensive, especially given the very high sampling rates and the requirement for

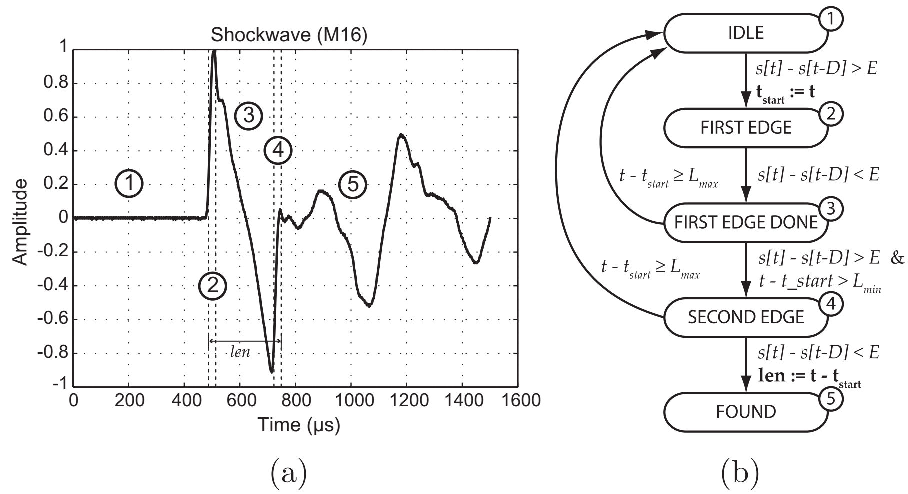


Fig. 4. (a) A representative shock wave signal generated by a 5.56 × 45 mm NATO projectile fired from a M16 rifle (b) and the state machine of the detection algorithm.

data parallelism. Second, due to the nonlinearities the analog channels exhibit under extremely high sound pressures (microphone saturation, oscillation at recovery, etc.), the frequency-domain image of the sampled signal may be quite different from that of the actual acoustic phenomenon.

Functionally, shockwave and muzzle blast detectors are implemented as independent IP cores within the FPGA: one pair for each of the four channels. The parameters of the detector cores are runtime-configurable through the virtual register file, and the register file is where the detection results (timestamps and event-specific feature vectors) are stored. While the operations of the shockwave and muzzle blast detection cores are independent, a crude local fusion module integrates them by shutting down those cores which missed their events after a reasonable timeout and by generating a single detection message to the mote. This is the point when the FPGA signals to the mote that detection data is available for downloading. After the mote sends back an acknowledgement, the detector cores reset and continue their operation.

6.1. Shockwave detection

The acoustic shockwave has a very characteristic signature in the time domain (see Fig. 4a). The signal shape resembles a capital letter "N", with steep rising edges at the beginning and end of the signal. The duration of the N-wave is (largely) determined by the caliber and miss distance (see Section 7.5), and it typically ranges between 200 and 300 μs for the configurations we consider.

The state machine of the shockwave identification algorithm is shown in Fig. 4b. Detection parameters, such as the minimum steepness of the edges (D,E) and bounds on the length of the wave ($L_{\min}, L_{\max}$), are runtime configurable through the register file. The output of the detector is the timestamp of the beginning of the shockwave, i.e. the time of the first sample of the first rising edge of the N-wave, and the shockwave duration. Both are written into the register file and can be accessed by the mote.

6.2. Muzzle blast detection

The acoustic signature of a muzzle blast can be characterized by a long initial period, ranging from 1 to 5 ms, where the first half of the period is significantly shorter than the second one [9]. Due to the physical limitations of the analog circuitry, as described at the beginning of this section, it is very possible that irregular oscillations and microphone saturation will show up within this longer time window. An example of this is shown in Fig. 5a. Because of this, the real challenge for the matching detection core is to properly identify the first and second half-periods properly.

The muzzle blast detector's state machine (Fig. 5b) does not work on the raw samples directly but is fed by a zero crossing (ZC) encoder. After the initial threshold-based triggering, the

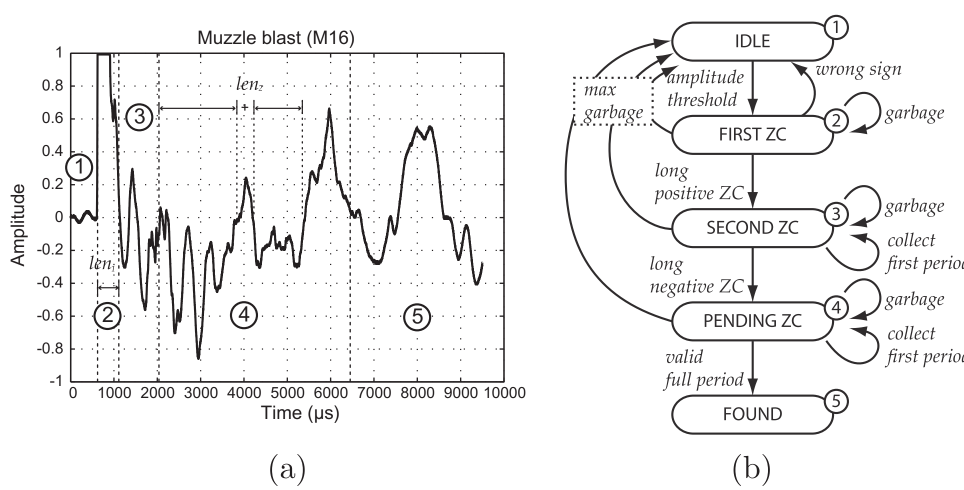


Fig. 5. (a) A representative muzzle blast signature produced by an M16 rifle and (b) the corresponding zero-crossing based (ZC) detection logic.

detector attempts to collect ZC segments belonging to the first period (positive amplitude) while discarding too short, possibly irregular segments, effectively implementing a rudimentary low-pass filter in the ZC domain. After a sufficiently long negative segment is detected, the algorithm runs the identical logic for the second half-period. If too many irregularities are observed in the collection phases, the detector resets itself to prevent the (false) detection of the halves from completely different periods separated by rapid oscillation or noise. Finally, if the detected muzzle blast candidate satisfies the constraints (length and ratio of period halves), the muzzle blast detector core writes into the register file the actual length, amplitude, and energy of the period calculated concurrently, and a detection event is signaled.

The initial triggering mechanism is based on two amplitude thresholds: a static one, the value of which is read from the register file, and a dynamic one. The dynamic threshold is essential to adapt the sensor to different ambient noise environments and to temporarily suspend the muzzle blast detector after a shockwave event. We observed that oscillations in the analog section or reverberations in the sensor enclosure can otherwise trigger false muzzle blast detections. The dynamic noise level is estimated by a single pole recursive low-pass filter (cutoff @ 0.5 kHz) on the FPGA.

7. Sensor fusion

The sensor fusion algorithm estimates shooter position, trajectory of the projectile, caliber, and weapon type using detection messages received from the sensor network. Computation consists of several distinct steps:

- Angle of arrival (AoA) computation of shockwave and muzzle blast for individual microphone arrays (see Section 7.1).
- Range calculation from local shockwave and AoA measurements (see Section 7.2).
- Trajectory computation from network-wide shockwave measurements (see Section 7.3).
- *If* trajectory is available: shooter position estimation using trajectory information (see Section 7.4), *otherwise*: shooter position estimation fusing network-wide muzzle blast AoAs and TDoAs, followed by a trajectory estimation based on the estimated shooter location and available network-wide shockwave AoAs and TDoAs (see Section 7.4).
- *If* trajectory is available: caliber estimation (see Section 7.5).
- *If* caliber is available: weapon classification (see Section 7.6).

7.1. AoA calculation

The sensor fusion algorithm first computes the shockwave and muzzle blast AoAs for each individual microphone array. Since the aperture of the individual arrays is small, we can safely rely on the far-field assumption that the wavefront impinging on the array is considered a planar wave.

First, we validate the time of arrival (ToA) measurements against constraints given by the geometry and the speed of sound: the TDoA measured between any two microphones cannot be larger than the sound propagation time between the two microphones. If this constraint is violated, we discard the corresponding detections and no AoA is computed.

Formally,

$$|t_i - t_j| <= |P_i - P_j|/c + \varepsilon$$

where P_1, P_2, and P_3 are the positions of the microphones ordered by time of arrival $t_1 < t_2 < t_3$, c is the speed of sound, and ε is a threshold on the maximum allowable detection error.

Once the detection values are validated, we compute the AoA using the following direct method: Let $v(x, y, z)$ be the normal vector of the unknown direction of arrival. We denote the vector from P_1 to P_2 with $r_1(x_1, y_1, z_1)$, the vector from P_1 to P_3 with $r_2(x_2, y_2, z_2)$, and the normal vector of the unknown AoA with $v(x, y, z)$. Consider the projection of the direction of motion of the wavefront (v) to r_1 divided by the speed of sound (c). This is exactly the time it takes for the wavefront to propagate from P_1 to P_2:

$$vr_1 = c(t_2 - t_1)$$

Similarly, for r_2 and v:

$$vr_2 = c(t_3 - t_1)$$

Given that v is a normal vector and rewriting the dot product in algebraic form, we get the following quadratic system:

$$xx_1 + yy_1 + zz_1 = c(t_2 - t_1)$$
$$xx_2 + yy_2 + zz_2 = c(t_3 - t_1)$$
$$x^2 + y^2 + z^2 = 1$$

To avoid verbosity, we omit the solution steps here, as they are straightforward. In the general case, there are two solutions, except when the source is on the $P_1P_2P_3$ plane and the two solutions coincide. To disambiguate between the solutions, we check which one aligns with the fourth measurement. Should there only be three measurements available, we forward both results to the subsequent computation steps.

7.2. Local shockwave/muzzle blast fusion

While the four muzzle blast detections would theoretically suffice for position estimation, the curvature of the spherical muzzle blast wavefront is negligible with the given small array aperture, making such an approach very sensitive to detection and quantization errors in the detection timestamps. Instead, we use an approach that makes use of the shockwave detections and the shockwave-muzzle blast TDoA to compute a location from measurements acquired on a single array. In a more general case, the shockwave and muzzle blast AoAs do not need to originate from the same array: this is the approach we describe below.

Consider Fig. 6. The shot was fired at time t from point P, both of which are unknown. Let us assume that a muzzle blast detector at position P_1 reported AoA u with timestamp t_1, and that a shockwave detector at position P_2 reported AoA v with timestamp t_2. We show that these measurements are sufficient to unambiguously compute the shooter position P.

Let P_2' be the point on the extended shockwave cone surface where PP_2' is perpendicular to the surface. Note that PP_2' is parallel to v. Since P_2' is on the cone surface which hits P_2, a sensor at P_2' would detect the same shockwave time of arrival (t_2). From the

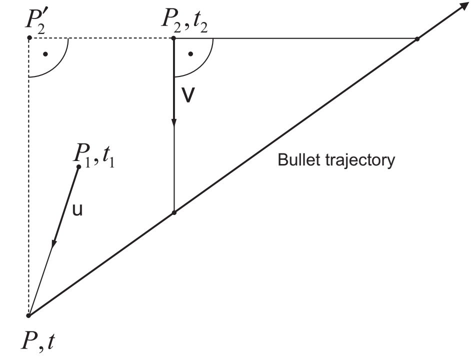


Fig. 6. Geometry of a shot and the corresponding trajectory, fired from point P at time t. Two sensors (at P_1 and at P_2) detect the shock wave and muzzle blast AoAs at times t_1 and t_2, respectively.

fact that the cone surface travels at the speed of sound c, we express P using P_2':

$$P = P_2' + cv(t_2 - t).$$

Similarly, P can be expressed from P_1:

$$P = P_1 + cu(t_1 - t)$$

yielding

$$P_1 + cu(t_1 - t) = P_2' + cv(t_2 - t).$$

P_2P_2' is perpendicular to v:

$$(P_2' - P_2)v = 0$$

yielding

$$(P_1 + cu(t_1 - t) - cv(t_2 - t) - P_2)v = 0$$

which contains only one unknown, t. Thus, we obtain:

$$t = \frac{\frac{(P_1-P_2)v}{c} + uvt_1 - t_2}{uv - 1}.$$

From here, we directly calculate the shooter position P.

In the special case where a single sensor detects both the shockwave and the muzzle blast AoAs, i.e. when $P_1 = P_2$, we get:

$$t = \frac{uvt_1 - t_2}{uv - 1}.$$

Note that even if the absolute orientation of the array is unknown, we can still get a good range estimate using t from this formula, because u and v are not used separately, only as uv.

This method has one caveat, namely that we assume the shockwave front to be a perfect cone. This, however, is not true in reality, because deceleration and the effects of gravity are present. This results in a wavefront resembling the pointed end of an American football. Deceleration causes the time difference between the shockwave and muzzle blast detections to decrease, because the shockwave generation slows down with the bullet. Since a smaller time difference results in a smaller range, the method we presented above underestimates the true range. Deceleration, however, is specific to the projectile, and as this method gave reasonable results for our purposes, we did not consider correction for deceleration in this project and leave this for future work.

7.3. Trajectory estimation

It is well known that trajectory and bullet speed can be computed with a direct formula from two shockwave AoAs and their TDoAs [6]. This technique suffers from high error sensitivity when the two shockwave directions are close to parallel. To avoid this, we discard shockwave AoA pairs when their angle is smaller than a predefined threshold.

The input set of shockwave AoAs contains incorrect values, because when only three microphones detect a shockwave, it is not possible to disambiguate the two solutions. Therefore, by generating all possible pairs of shockwave AoAs that satisfy the angle threshold constraint, we end up computing a large number of trajectory candidates with Danicki's method, a significant portion of which are incorrect. We use outlier filtering and averaging of the trajectory candidates to find the trajectory estimate.

Filtering is carried out as follows. We define a distance function for a pair of trajectory candidates as the scalar product of their normal vectors:

$$D(i,j) = v_i v_j$$

Then, we define a neighbor set for each trajectory candidate:

$$N(i) := \{j | D(i,j) < R\}$$

where R is a predefined radius parameter. We consider the neighbor set with the highest cardinality as the core set C containing the real solution; all other trajectories are outliers. The core set can be found in $O(N^2)$ time, assuming that all $N(N-1)$ AoA pairs are used as inputs (N sensors all reporting two ambiguous AoAs, all pairs being over the angular threshold). The trajectory candidates in C are averaged to get the final trajectory.

Note that there is a common geometric configuration where no shockwave AoA pairs meet the angular threshold constraint. This is the case when all the sensors that detect the shockwave AoA are on one side of the trajectory, and the trajectory is roughly in the same plain as the sensors are deployed. This happened multiple times during the evaluation of the system, and can be addressed by first computing the shooter position (see Section 7.4). Assuming that a point on a trajectory is known, it is sufficient to compute the speed of the projectile v to get the trajectory estimate. We employ an exhaustive high resolution search to find v in this case, minimizing an error function defined over the shockwave AoAs.

7.4. Shooter position estimation

We have several classes of information (measurements and derived values) that help us find the shooter location:

- Shooter position estimates reported by individual sensors that detect both shockwave and muzzle blast AoA.
- Shooter position estimates computed from two distinct arrays detecting a shockwave and a muzzle blast AoA, respectively.
- Muzzle blast AoAs, which, in effect, give a bearing estimate from the corresponding array to the shooter, constraining the shooter position on a half line.
- Muzzle blast ToAs, four of which are sufficient to locate the shooter with multilateration (in the general case).

While these measurements and derived values all have different error characteristics, all of them are susceptible to acoustic echoes, which cause the values to be incorrect. Also, some derived values, e.g. the shooter position computed from shockwave and muzzle blast AoAs, may be incorrect due to ambiguous results being reported (e.g. refer to Fig. 7). Therefore, standard nonlinear optimization techniques (e.g. Newton–Raphson, etc.) fail due to the potentially large number of outliers.

We have shown in our earlier work that a similar problem can be solved efficiently with an interval arithmetic-based bisection search algorithm [12]. We define a consistency function for a box in three dimensional space as the number of measurements or derived values, weighted differently for each information class, that are consistent with the hypothesis that the given box contains the shooter's location. The search starts with a box large enough to contain the entire area of interest, then iterates by dividing and evaluating boxes. The box with the maximum consistency is divided until the desired precision is reached. Backtracking is possible to avoid getting stuck at a local maximum.

Let us first consider the case where no trajectory information is available. For a box B, we use the available information as follows. ToA estimates at different microphones from a single array are averaged to use a single ToA per node. Then, for each ToA, we can calculate the corresponding time of the shot that is consistent with the given box. Since the shooter is assumed to be somewhere inside the box B, which is not a single point, this gives us an interval for the shot time. The maximum number of overlapping time intervals gives us the value to contribute to the consistency function for B. This approach is identical to what we used in [12]. Similarly, if the positions reported by individual nodes or pairs of nodes are within box B, they increase the value of the consistency function. Analogously, the value of the consistency function for box

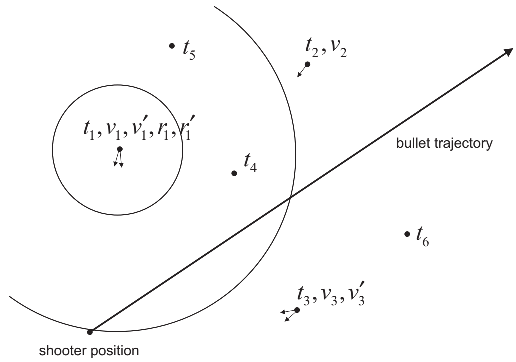


Fig. 7. Example of heterogenous input data for the shooter position estimation algorithm. All sensors have ToA measurements (t_1, t_2, t_3, t_4, t_5), one sensor has a single bearing estimate (v_2), one sensor has two possible bearings (v_3, v_3') and one sensor has two bearing and two range estimates (v_1, v_1', r_1, r_1').

B is increased proportionally to the number of muzzle blast AoAs when the corresponding line segments intersect the box. We apply empirically derived weights to balance the contribution of the different information classes to the consistency value.

If a trajectory estimate is available, the shooter location is constrained to a line, so the above algorithm can be used along the estimated trajectory, resulting in several orders of magnitude improvement in computation time.

7.5. Caliber estimation

Whitham showed that the shockwave period T is a function of the projectile diameter d, the length l, the distance b of the microphone from the trajectory of the bullet, the Mach number M, and the speed of sound c [22]:

$$T = \frac{1.82\ \mathrm{Mb}^{1/4}}{c(M^2-1)^{3/8}}\frac{d}{l^{1/4}} \approx \frac{1.82d}{c}\left(\frac{\mathrm{Mb}}{l}\right)^{1/4}$$

We investigated the relationship between miss distance and shockwave length experimentally. In the experimental data set, 196 shots were fired with three different bullet calibers. A stationary deployment of ten sensors, each equipped with four microphones, was used to measure the shockwave lengths. We filtered the results such that measurements from nodes where at least three out of the four microphones did not agree on a shockwave length to within at most 5 µs were discarded. This filtering lead to an average per-sensor detection rate of 86%. The collected data is shown in Fig. 8, plotting measured shockwave lengths versus the miss distance. Whitham's formula suggests that the shockwave length for a given caliber can be approximated with a power function of the miss distance (with a 1/4 exponent). Best fit functions on our data are:

$$.50\ \mathrm{cal}: T = 237.75b^{0.2059}$$
$$7.62\ \mathrm{mm}: T = 178.11b^{0.1996}$$
$$5.56\ \mathrm{mm}: T = 144.39b^{0.1757}$$

These values were derived by minimizing the RMS error of the filtered sensor readings. When the above formulas were used to estimate caliber, caliber estimation was correct in more than 99% of the shots within this data set (when a trajectory estimate was available). Note, however, that caliber estimation works only if there are enough shockwave detections to compute the trajectory.

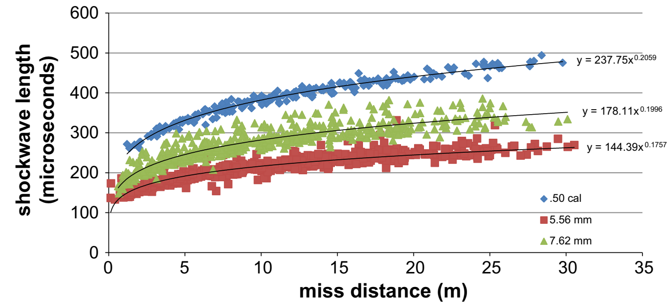


Fig. 8. Relationship between shockwave length and miss distance. Each data point represents one sensor board after an aggregation of the individual measurements of the four acoustic channels. Three different caliber projectiles have been tested (196 shots, 10 sensors).

7.6. Weapon classification

We carried out several measurements in urban-like, high-multipath environments to identify weapon-specific characteristics in the muzzle blast's acoustic signature without much success. We concluded that the signal shape of the muzzle blast is not sufficiently characteristic of a particular weapon; therefore, we did not find it usable for classification purposes. We suspect that reverberation and echoes of the high energy muzzle blast from the environment have a much higher impact on the muzzle blast signal shape than the weapon itself.

However, the observed speed of the projectile correlated well with the weapon type. The reason for this is that the muzzle speed of a given weapon is nearly constant, assuming that the same type of ammunition is used. Figs. 9 and 10 present the relationship between range and measured bullet speed for different calibers and weapons. When supersonic projectiles are considered, deceleration of the bullet can be approximated by a cubic function, with a y-intercept value equal to the muzzle velocity of the weapon–ammunition pair. Such a deceleration function aligns well with the dynamics of the physical deceleration phenomena due to air resistance. It is immediately apparent from Fig. 9 that two of the weapons tested with 7.62 mm caliber (AK47, M240) ammunition can be clearly distinguished from each other. Unfortunately, this does not always hold true in the 5.56 mm caliber case: the M16 with its higher muzzle speed can still be well classified, but the M4 and M249 have similar muzzle velocities (see Fig. 10); therefore, it is not always possible to unambiguously classify them. The lower weapon classification success rate for these two weapons can be attributed to the limited number of practice shots we were able to take before actual testing began, which would have provided a more extensive training data set, which, in turn, could have revealed a better separation between the two weapons.

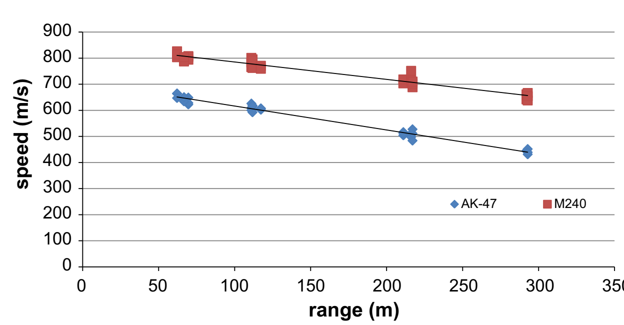


Fig. 9. AK-47 and M240 bullet deceleration measurements. Both weapons have the same caliber. Data is approximated using simple linear regression.

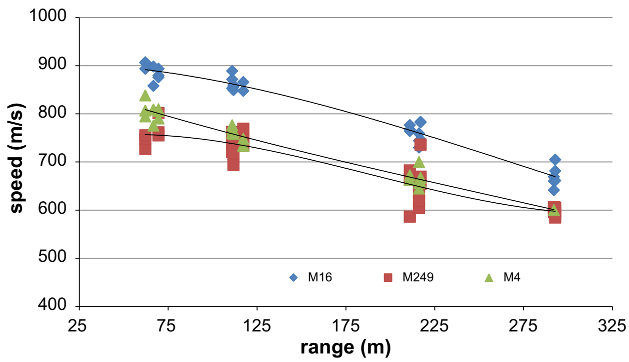


Fig. 10. M16, M249 and M4 bullet deceleration measurements. All weapons have the same caliber. Data is approximated using simple cubic regression.

Weapon classification is carried out as follows. If a trajectory estimate is available, the bullet velocity along the trajectory can be estimated based on sensor locations and the corresponding shockwave AoAs. From the estimated shooter position, the various muzzle speeds and deceleration functions provide bullet speed estimates at each sensor location. To classify the weapon for a given a shot, we choose the weapon type with the smallest RMS error with respect to the measured projectile speed values.

8. Standalone operation

Even without network connectivity, the wearable microphone array can still provide utility to the warfighter. If there are at least three microphones that detect the muzzle blast, the AoA yields the direction to the shooter (see Section 7.1). If both muzzle blast and shockwave AoAs are available, the shooter location can be calculated locally (see Section 7.4).

Interestingly, the system can still provide the soldier with useful information when the number of detections is not sufficient to compute the shooter's location or bearing. Assuming a given caliber (e.g. a 7.62 mm projectile), the miss distance can be estimated using the length of a single shockwave detection using the relationship between the two quantities described in Section 7.5.

When there is at least one muzzle blast and one shockwave detection on the same node, it is possible to compute a reasonable estimate of the shooter's range, relying on assumptions about the caliber (e.g. 7.62 mm) and approximate muzzle speed (e.g. 710 m/s for a typical AK-47). Note, it is not necessary for the same microphone to make both detections as the microphone separation in the array is negligible compared to the typical range of the shooter. On the other hand, this also means that a single-channel acoustic sensor can determine the miss distance and the shooter range based on a single shot.

Consider Fig. 11. Points S and M represent the locations of the shooter and the microphone detecting the acoustic events, respectively. Let us denote the range, i.e. the Euclidean distance between points S and M, with $d_{S,M}$. Assuming line-of-sight (LOS) conditions, the detection time of the muzzle blast can be written as

$$t_{mb} = t_{shot} + \frac{d_{M,S}}{c} \tag{1}$$

where t_{shot} is the time of shot, and c is the speed of sound which is assumed to be known. That is, the muzzle blast travels at the speed of sound from S to M, and is, therefore, detected $\frac{d_{M,S}}{c}$ seconds after the weapon is fired.

Since the shockwave does not originate from the muzzle but from the bullet traveling along its trajectory, we need to consider the time traveled by the bullet from the muzzle to a point P on the trajectory, as well as the time traveled by the wavefront of

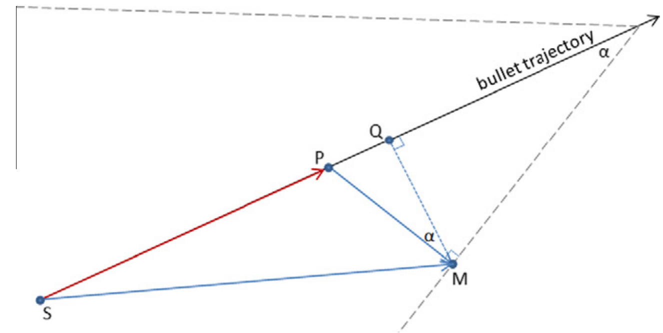


Fig. 11. Calculation of range from miss distance and shockwave-muzzle blast TDoA.

the shockwave from point P to the microphone. For simplicity, let us assume for now that the bullet travels at a known constant speed v. The shockwave detection time can be written as

$$t_{sw} = t_{shot} + \frac{d_{S,P}}{v} + \frac{d_{P,M}}{c} \tag{2}$$

The measured shockwave-muzzle blast TDoA can be expressed as

$$t_{mb} - t_{sw} = \frac{d_{M,S}}{c} - \frac{d_{S,P}}{v} - \frac{d_{P,M}}{c} \tag{3}$$

Since we assume a constant bullet speed v, the shockwave front has a conical shape, such that the angle between the trajectory and the conical surface is

$$\alpha = sin^{-1}\frac{c}{v} \tag{4}$$

We notice that $\measuredangle PMT$ is a right angle; therefore, $\measuredangle PMQ = \alpha$ because the respective triangles are similar. Knowing α and the calculated miss distance $d_{Q,M}$, we can express distances $d_{S,P}$ and $d_{P,M}$ as follows:

$$d_{P,M} = \frac{d_{Q,M}}{cos(\alpha)} \tag{5}$$

and

$$d_{S,P} = d_{S,Q} - d_{P,Q} \tag{6}$$

where

$$d_{S,Q} = \sqrt{\left(d_{S,M}^2 - d_{Q,M}^2\right)} \tag{7}$$

using the Pythagorean theorem, and

$$d_{P,Q} = d_{Q,M} tan(\alpha) \tag{8}$$

Therefore, Eq. (3) can be solved for the range $d_{S,M}$, resulting in a closed-form formula that is extremely fast to evaluate on the target hardware:

$$d_{S,M} = \frac{1}{2(c^4 - v^4)}(P - 2\sqrt{Q}) \tag{9}$$

where P and Q are defined as

$$\begin{aligned} P = {} & -2v^3 d_{Q,M}\sqrt{v^2 + c^2} - 2(t_{mb} - t_{sw})c^3 v^2 + 2c^2 d_{Q,M} v\sqrt{v^2 + c^2} \\ & - 2(t_{mb} - t_{sw})c v^4 \\ Q = {} & -2c^4 v^4 d_{Q,M}^2 + 2(t_{mb} - t_{sw})^2 c^6 v^4 \\ & + (t_{mb} - t_{sw})^2 c^4 v^6 - 2c^7 d_{Q,M}(t_{mb} - t_{sw}) v\sqrt{v^2 + c^2} \\ & + c^8 (t_{mb} - t_{sw})^2 v^2 + 2c^8 d_{Q,M}^2 + 2v^5 d_{Q,M}\sqrt{v^2 + c^2}(t_{mb} - t_{sw})c^3 \end{aligned}$$

While relaxing the assumption of constant bullet speed and incorporating a weapon-specific deceleration constant in the equations would result in more precise results, finding the solution for $d_{M,S}$ would require numerical methods, making the algorithm more computationally expensive.

9. Results

A team from NIST carried out an evaluation of our shooter location system at the US Army Aberdeen Test Center in April 2006 [21]. The setup emulated an urban environment with mock-up wooden buildings to create reverberations and echoes. We deployed 10 sensor nodes at surveyed locations in an approximately 30×30 m area. The sensor positions remained static during the evaluation, primarily for safety reasons; i.e. the mobile aspects of the system were not tested. Five fixed targets behind the sensor network were used for aiming. Firing positions were located at 50, 100, 200 and 300 m away from the targets. These positions were known to the evaluators, but not to the operators of the system. Six different weapons types were tested: AK-47 and M240 firing 7.62 mm projectiles, M16, M4 and M249 with 5.56 mm ammunition, and the .50 caliber M107.

9.0.1. Detection rates

Table 1 summarizes the evaluation results. One hundred ninty-six shots were fired throughout the test, all of which were detected by the system. Our shockwave detector logic was proven to be robust; there were no false shockwave detections. However, we observed a number of false muzzle blast detections, which could be attributed to artillery fire at a nearby shooting range.

9.0.2. Localization rates

All but seven of the 196 shots fired were localized. In the case of all seven missed shots, the bullet passed on one side of the sensor field, which resulted in nearly identical shockwave AoA estimates for all sensors. Because the shockwave AoAs are subject to threshold-based filtering to eliminate bearings that are close to parallel (to avoid excessive error amplification), these geometries cannot leverage our robust trajectory estimator. In this scenario, two significantly different trajectories can generate the same set of observations (see [12] and also Section 7.3). Instead of estimating which one is more likely or displaying both possibilities, we chose not to report a trajectory at all.

We improved on the localization algorithm after the evaluation, implementing the method described in Section 7.3. Utilizing the log of detection results collected during the evaluation, we were able to localize 5 of the 7 shots we missed during the test. This enhancement improved the localization rate of the system to 99% (see Fig. 12).

9.0.3. Localization accuracy

Localization accuracy is broken down into accuracy of trajectory angle, accuracy of translation, and range estimation precision. We observed that trajectory accuracy does not depend on the range, as long as the bullet passes over the network. The angular error of trajectory estimation was under 1 degree for all shots except a few fired from a distance of 300 m. We suspect that the slightly less accurate trajectory angles for the 300 m range can be attributed to human error, as these shots were fired in a hurry near the end of the day. This claim is also supported by the logged sensory data: the estimated trajectory distance from the actual targets has an average error of 1.3 m for 300 m shots, 0.75 m for 200 m shots, and 0.6 m for all other shots. As the distance between the targets and the sensor network was fixed, this number should not show a 2× improvement just because the shooter is closer. This suggests that longer range shots were less accurate; that is, the presumed ground truth on the actual trajectories must be inaccurate.

Translational trajectory error (see Table 1) refers to the distance of the actual shooter position from the estimated trajectory. Typical translational errors are 1–2% of the actual range. Intuitively, the farther the shooter is from the network, the larger the translational errors are expected to be. This justifies the use of expressing translational errors as a percentage of the range. Again, we believe that the disproportionately larger errors at 300 m were due to human error in aiming. When computing translational errors, we assumed that the precise location of the shooter and the target were known. Since target locations are imperfect for the 300 m shots, any inaccuracy in the actual trajectory directly adds to the calculated error, which explains why the results appear weaker.

9.1. Range estimation

In our nomenclature, the term *range estimation* refers to finding the shooter position on the calculated trajectory. While range estimation traditionally refers to finding the distance between the sensor and the shooter, in our scenario, we pick a point on the trajectory estimate inside the sensor network as the origin and measure the distance to the shooter location from there. With this notion of range, the range estimates computed by the system are better than 5% accurate from 50 m, 10% for 100 m, and 20% or worse for longer distances.

One problem we faced during the field test was that the sensors sometimes mistakenly characterized shockwave echoes as muzzle blasts. For long range shots, echoes of the shockwave reach the sensors earlier than the muzzle blast, since the projectile travels 2–3× faster than the speed of sound, so the time between the shockwave (and its possible reverberations) and the muzzle blast increases with increasing ranges. As a result of these false muzzle blast detections, the range was often underestimated, as the system measured shorter times than actually occurred.

We improved on the muzzle blast detection algorithm after the evaluation; however, we have no means of telling how these enhancements would improve the evaluation results, as we only logged detection times and signal features, not the raw acoustic samples which are the inputs to the muzzle blast detector.

9.1.1. Weapon classification

We analyze the accuracy of our caliber estimation and weapon classification technique based on the 194 shots that were success-

Table 1
Summary of results fusing all available sensor observations. All shots were successfully detected, so the detection rate is omitted. Localization rate means the percentage of shots that the sensor fusion was able to estimate the trajectory of. The caliber accuracy rate is relative to the shots localized, not the total number of shots since caliber estimation requires trajectory. The trajectory error is broken down to azimuth in degrees and the actual distance of the shooter from the trajectory. The distance error shows the distance between the real shooter position and the estimated shooter position. As such, it includes the error caused by both the trajectory and that of the range estimation. Note that the traditional bearing and range metrics are not good for a distributed system such as ours because of the lack of a single reference point.

Shooter range (m)	Localization rate (%)	Caliber accuracy (%)	Trajectory Azimuth error (deg)	Trajectory distance error (m)	Distance error (m)	No. of shots
50	93	100	0.86	0.91	2.2	54
100	100	100	0.66	1.34	8.7	54
200	96	100	0.74	2.71	32.8	54
300	97	97	1.49	6.29	70.6	34
All	96	99.5	0.88	2.47	23.0	196

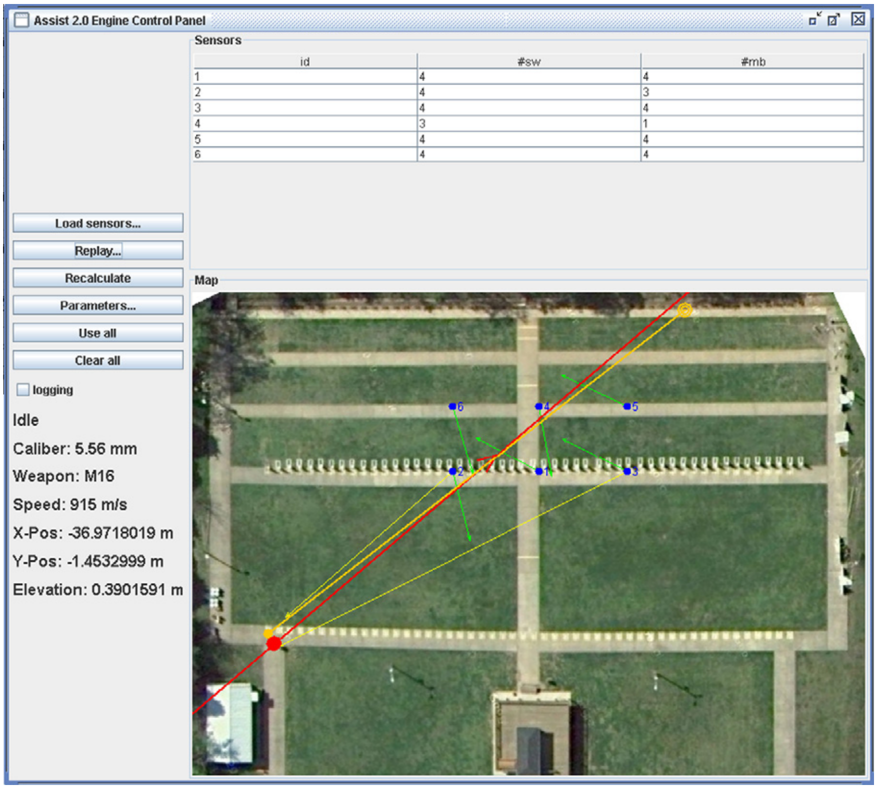


Fig. 12. The user interface of the system showing an experimental setup. Sensor nodes are labeled by their node IDs and marked by blue dots. The shooter position is marked by an orange dot, the target by a bull's eye, and the trajectory by an orange line. The long yellow arrows point to the shooter position estimated by each sensor. Where it is missing, the corresponding sensor did not have enough detections to measure the AoA of either the muzzle blast, the shockwave or both. The shorter green arrows represent the shockwave AoA detections. The red line and large dot represent the estimated trajectory and shooter position, respectively. This particular shot was fired at a shooting range in Nashville, TN, the shooter being approximately 40 m away from the sensor field. Both trajectory and shooter position were localized almost perfectly by the sensor network. The caliber and weapon were also identified correctly. Two out of six nodes were able to estimate the location alone. Their bearing accuracy is within a degree, while the range is off by less than 5%. (For interpretation of the references in colour in this figure legend, the reader is referred to the web version of this article.)

fully localized with the post-evaluation, improved version of our system. The caliber estimator correctly identified the bullet caliber for all but one of the 194 shots. The overall weapon classification accuracy of 86% might not seem too impressive, but when we look at the results broken down by the type of weapon used, as shown in Table 2, the picture becomes more subtle. For four of the weapons (AK-47, M16, M240 and M107), only one shot out of approximately 140 was erroneously classified.

However, since the M4 and the M249 use the same caliber bullet and their muzzle speeds are very similar, it was not possible to correctly disambiguate them 100% of the time. Unfortunately, we only took a small number of test shots before the evaluation that were used to compute the deceleration parameters. We believe that it should be possible to create more accurate deceleration approximation functions when more calibration data is available, which could significantly improve the accuracy of our weapon classification algorithm. Note that when the M4 and the M249 are considered a single weapon class, overall classification accuracy jumps to 98%.

Apart from disambiguating the two aforementioned weapon types, the system is able to classify weapons that have the same caliber. For instance, the AK-47 and M240 have the same caliber (7.62 mm), just as the M16, M4 and M249 do (5.56 mm). This is an important contribution of our work. We are not aware of any system that classifies weapons with this accuracy.

9.2. Single sensor performance

As described in Section 7.2, even a single sensor can localize the shooter if both shockwave and muzzle blast AoAs and their corresponding detection times are available. This approach, however, does not work for shooter-sensor ranges larger than approximately 150 m, because our sensors are unable to pick up or accurately timestamp muzzle blast signatures from such long distances. Because of this, we only evaluate the single sensor localization accuracy for the 108 shots that were taken from closer than 150 m. Note that we use the same test data as in the previous section, but we evaluate individually for each sensor.

Bearing and range estimation accuracy for individual sensors is presented in Table 3. By bearing error, we mean the difference between the muzzle blast AoA reported by the sensor and the angle to the shooter position in the sensor's coordinate system. Range refers to the Euclidean sensor-shooter distance. Since these are not networked benchmarks, results are free from time-synchronization errors that would manifest only in a networked system. Also, in the case of range estimation results, the orientation errors of the sensors are irrelevant. While position or bearing estimation by a single sensor is not as robust as networked sensor fusion (nor is it possible to carry out caliber estimation or weapon classification in the single sensor scenario), these benchmarks are still important to gain a better picture of the accuracy of the individual building blocks of our system.

Table 3 indicates that there is a large variance in the performance of individual sensors. This, in part, has to do with the particular geometric configuration of the sensors, shooter and target positions, obstructions of line of sight, and exposure to possible echoes. Also, the sensors were hand-built prototypes, employing consumer-quality microphones with mechanical damping. Similarly, neither packaging nor mounting was production-quality. Still, we consider the average bearing error of 0.9 degrees and

Table 2
Weapon classification results. The percentages are relative to the number of shots localized, not the total number of shots, as the classification algorithm needs to know the trajectory and the range. Note that the difference is small; 194 shots were localized out of the total 196.

Distance (m)	M16 (%)	AK47 (%)	M240 (%)	M107 (%)	M4 (%)	M249 (%)	M4-M249 (%)
50	100	100	100	100	56	88	100
100	100	100	100	100	56	44	100
200	89	100	100	100	89	33	100
300	100	100	100	100	50	33	80
All	97	100	100	100	66	52	98

range error of 5 m very promising. Our opinion is that accuracy could be greatly improved through professional manufacturing and the use of custom microphones.

The percentage of nodes that yielded localization results was 42%. If only 50 m shots are considered, the localization rate improves to 61%. We estimate that about one tenth of the sensors did not have unobstructed line of sight to the shooter position. Since local position calculation requires both shockwave and muzzle blast AoA detections, it is impossible, due to the physical geometry, for these sensors to accurately locate the shooter. However, if we consider only those sensors that picked up at least one shockwave, the detection rate is practically 100%.

The largest observed range error was 90 m, probably due to a shockwave echo incorrectly detected as a muzzle blast, and the largest bearing error was approximately 12 degrees. Even with these errors, this is a valuable piece of information for a soldier about the location of an enemy.

If the number of shockwave and muzzle blast detections at a particular sensor is not sufficient to localize the shooter, it is still often possible to estimate the miss distance. The networked operating mode of the system uses this relationship to estimate the caliber of the projectile based on the known miss distance that was available due to very precise trajectory estimation (using the relationship between the two quantities described in Section 7.5). In the standalone mode of operation, we turn the problem around: assuming that the caliber is 7.62 mm, what is the corresponding miss distance? Since the weapon the enemy uses is commonly known, this assumption is not a limiting factor in practice. What kind of accuracy in miss distance estimation can we then expect?

From the experimental live fire detection data set, we selected all 166 shockwave detections of AK-47 shots fired from a distance of 50–130 m from the sensors, and computed the miss distance estimates using the above formula. The results, actual vs. estimated miss distances, are shown in Fig. 13. The actual miss distances ranged from 0 to 27.5 m.

As expected, the accuracy of miss distance estimation degrades as the miss distance grows. The reason for this is inherent in how the shockwave length is measured. Close to the bullet's path, the shockwave has a very pronounced 'N' shape, with sharp rising and falling edges. The beginning of the sharp edges can be identified relatively unambiguously using a threshold on the slope. However, when the sensor is far from the trajectory, the rising and falling edges are smoother and less pronounced, which makes it harder for our detection algorithm to unambiguously identify when the 'N' wave starts and ends. This uncertainty propagates into the miss distance estimates. Still, the average absolute error of miss distance estimation was 1.2 m.

As described in Section 8, it is possible to compute the range when there is at least one shockwave and muzzle blast detection available on a single node. This range calculation technique has two main sources of error: inaccuracy of the miss distance estimate and the assumption of a fixed projectile velocity. However, this can be greatly improved. Since the weapon-specific bullet deceleration function is known (see Figs. 9 and 10), we can use a simple iterative approach. Initially the range is estimated using a projectile speed of 650 m/s assuming an AK-47 from approximately 50 m range. Once the actual range estimate is computed, the speed corresponding to that range is used to compute a new range estimate.

We selected all 166 detections of AK-47 shots from different ranges between 50 and 130 m. We computed the miss distances as described in Section 7.5. Then, we computed the range as described above, assuming a projectile speed of 650 m/s and sound propagation speed of 340 m/s. When the resulting ranges were 50 m to 100 m, 100 m to 150 m, and 150 m to 200 m, respectively, we recalculated the range with an adjusted projectile speed assuming 312 m/s^2 deceleration, obtained empirically from the ASSIST data. The resulting range estimates versus the ground truth ranges are shown in Fig. 14. Discarding the two outliers, the average absolute estimation error is 5.4% of the actual range.

9.3. Error sources

Several factors contribute to shooter localization, trajectory estimation, and weapon classification errors, including inaccuracy of time synchronization in the sensor network, positioning and orientation errors of the sensors, incorrect estimate of the speed of sound, as well as the geometry of the sensor field, shooter position, and trajectory of the projectile.

We use routing-integrated time synchronization [17] within the sensor network, which relies on MAC-level timestamping. On the MICAz mote, timestamping has accuracy in the 10 µs range per radio hop. While this error is additive in a multihop scenario, we estimate the maximum synchronization error in our 10-node deployment to be around 100 µs. Since sound travels about 34 mm in that time, it is safe to consider time synchronization errors negligible.

Errors in node location and orientation, however, play an important role in overall system performance. To gain an insight into the effects of imprecise sensor positioning, we ran the sensor fusion algorithm on the sensory data collected during the test in Aberdeen, but added error terms to the surveyed node positions and orientations. We also tested the system with a reduced number of detections to see how the system performs when fewer sensors report shockwave and muzzle blast detections.

Table 3
Single sensor accuracy for 108 shots fired from 50 and 100 m. Localization rate refers to the percentage of shots the given sensor alone was able to localize. The bearing and range values are average errors. They characterize the accuracy of localization from the given sensor's perspective.

Sensor ID	1	2	3	5	7	8	9	10	11	12
Loc. rate (%)	44	37	53	52	19	63	51	31	23	44
Bearing (deg)	0.80	1.25	0.60	0.85	1.02	0.92	0.73	0.71	1.28	1.44
Range (m)	3.2	6.1	4.4	4.7	4.6	4.6	4.1	5.2	4.8	8.2

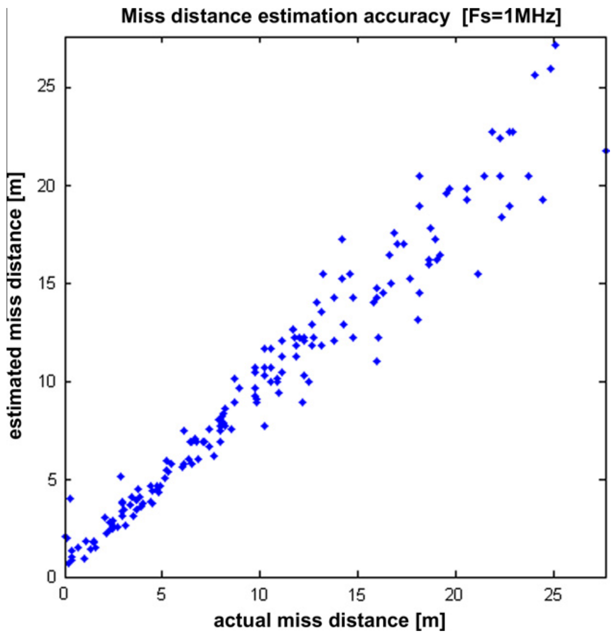


Fig. 13. Actual vs. estimated miss distances.

We ran four simulations, each of them one hundred times, where the number of sensors reporting detections was varied from 2 to 8 in increments of 2. Participating sensors were picked randomly in each simulation round. For each new round, the sensor location was perturbed by adding a two-dimensional zero-mean Gaussian random vector, effectively moving the sensor to a new random position around its real location. Similarly, the orientations were modified with an additive zero-mean Gaussian noise in each new round.

Fig. 15 presents the simulation results, one three-dimensional bar chart for each of the four simulations using 2, 4, 6, and 8 sensors, respectively.

The standard deviation of the Gaussian distribution used to perturb the node locations is displayed on the x-axes, ranging from 0 to 6 m. The y-axes displays the standard deviation of the Gaussian

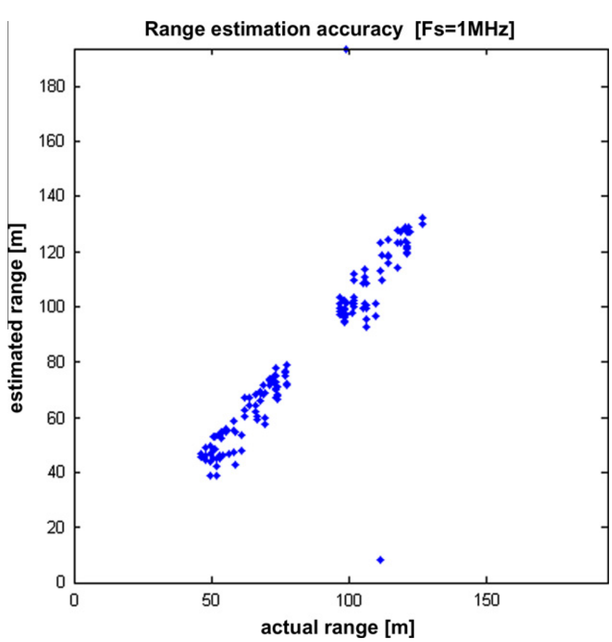


Fig. 14. Actual vs. estimated ranges.

error added to node orientations, ranging from 0 to 6 degrees. The height of the bars represent the angular trajectory errors.

Interestingly, node location errors of this magnitude have only a minor effect on the overall sensor fusion errors. On the other hand, system performance quickly degrades when orientation errors occur. The largest angular trajectory errors of 3.5° are observable in the two-sensor case, as well as in the six-sensor setup with highly perturbed orientations. It is important to note that, although the system works in three dimensions, there is minimal variance in the z-coordinates of the sensors; therefore, the overall geometry has insufficient vertical resolution. Because of this, the elevation angles were subject to larger errors than the azimuths.

While trajectory estimation accuracy does not significantly degrade with noisy sensor positions or with lower number of detections, the localization rate does suffer. That is, if the inputs are few or very inconsistent, the system is less likely to report a result, which is a desirable property. As we can see in Table 4, the number of successful localizations decreases from almost 100% to 50% when the number of sensors goes down from 8 to 2, even when no added noise is present in the detections. This can be primarily attributed to the geometry: system performance significantly degrades when the trajectory is on one side of the sensor network, which, obviously, is more likely to happen when fewer sensors are used. This also implies that the success rate of trajectory estimation (and therefore that of shooter localization and weapon classification) will suffer even when there are many sensors if they are located close to one another. On the other hand, just a few precise observations will result in accurate results if there is sufficient spatial separation between the sensors detecting the acoustic features. The simulation results also support the observation that success rate improves with the number sensors reporting detections: when 8 sensors are used, the localization rate was mostly independent from the magnitude of the synthetic errors.

Simulation results suggest that increasing sensor density beyond a point yields little improvement in localization rate; there was little difference in the results when more than 4 sensors were used. Therefore, we conclude that 6 sensors are sufficient for covering the 30 by 30 m area in our test scenario.

There are several other issues that may cause performance degradation. The sensor fusion algorithm needs to know the speed of sound, which depends primarily on the ambient temperature, as well as on relative humidity, the amount of CO_2 in the air, etc. We considered it a constant in the prototype system, which was set before testing. For a production system, it would be straightforward to either periodically download weather data from the internet or employ local temperature and humidity sensing to maintain a correct speed of sound estimate.

Wind also has an adverse effect on system performance as it alters the trajectory, as well as sound propagation. While it is possible to compensate for these effects, it is beyond the scope our work.

Finally, we present a few notes on the overall applicability of the system. Shockwave detections play an essential part in the sensor fusion. For subsonic projectiles (which is often the case for handguns), this approach will not work, since there will be no shockwave present. Also, since muzzle blast detections are important for shooter position estimation and weapon classification (but not for trajectory estimation), only the trajectory will be estimated when no muzzle blasts are observed (e.g. a silencer is used).

Although we did not test the variation of weapon classification parameters when different types of ammunition are used in the same weapon, we suspect that the classifier will have to be trained for different classes of ammunition for each supported weapon. This a crucial issue in a production system, since irregular armies may use substandard, even hand-manufactured bullets, potentially causing an insufficiently calibrated classifier to fail under such circumstances.

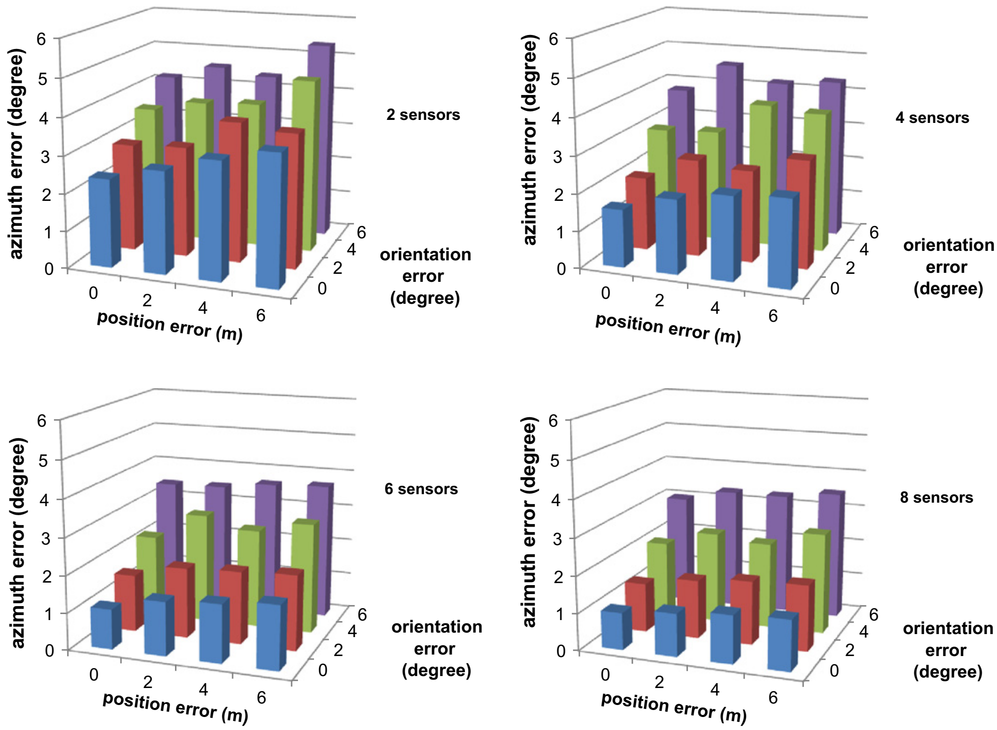


Fig. 15. The effect of node localization and orientation errors on azimuth accuracy with 2, 4, 6, and 8 nodes.

Table 4
Localization rate as a function of the number of sensors used, the sensor node location, and orientation errors. This simulation used the original approach utilized during the test and not the improved technique.

Errors/sensors	2 (%)	4 (%)	6 (%)	8 (%)
0 m, 0 deg	54	87	94	95
2 m, 2 deg	53	80	91	96
6 m, 0 deg	43	79	88	94
0 m, 6 deg	44	78	90	93
6 m, 6 deg	41	73	85	89

10. Conclusions

Our prototype system is characterized by high accuracy. A one-degree trajectory angle precision is unprecedented. Over 95% caliber estimation accuracy and close to 100% success on weapon classifications for 4 of the 6 weapons tested is also excellent. The simulation results utilizing real field data indicate that the system is robust against node location and orientation errors. The key factor behind this is the inherent outlier rejection capability of the sensor fusion algorithm. In this paper, we have showed significant improvements in the localization rate and the classification accuracy for the two weapons that fared poorly during the initial test.

Single sensor performance is also exceptional. The average bearing error of less than a degree and the typical 5% range precision are remarkable using a tiny microphone array. While the overall per-sensor localization rate for shots up to 130 m was only 42%, 87% of these shots were successfully localized by at least one of the ten sensors in standalone mode.

Finally, we have demonstrated that it is possible to accurately estimate the range and miss distance of a single shot using a single microphone. By measuring the shockwave length and the time difference of arrival between the shockwave and muzzle blast, and assuming a known weapon classification, an analytical formula provides both estimates with remarkable accuracy. Note that the weapon of choice of the overwhelming majority of fighters that NATO forces face today is the AK-47. Hence, this assumption is not limiting in practice.

There are significant further challenges before our system can be deployed. Power management is always important in battery-operated devices. However, standard duty cycling techniques would not work in this case without missing the first shot. The mobility aspect of the system has never been tested due to safety and the corresponding logistical issues. It is not clear that the sensor location and orientation accuracy that currently-available technologies provide is sufficient and/or cost effective. Other relatively minor outstanding issues include weatherproof packaging and ruggedization, as well as integration with current military infrastructure.

Acknowledgments

The Darpa IpTO ASSIST program, the US Navy through STTR N06-T004, Puritan Research Corporation and ARO MURI Grant W911NF-06-1-0076 have supported this research. We'd like to express our gratitude to Brian A. Weiss, Craig Schlenoff and the US Army Aberdeen Test Center who carried out the independent evaluation of the system. We are also grateful to Miklós Maróti, Gyula Simon, Branislav Kusý, Béla Fehér, Sebestyén Dóra, Ken Pence, Ted Bapty, Jason Scott, Christopher Nash, Dan Siewiorek and the Nashville Police Academy for their comments and contributions.

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